

PYTHON PROGRAMMING ASSESSMENT

Topics: List Slicing, List Methods, and Element Removal Operations

20 Objective Multiple-Choice Questions

1. Given the list `nums = [10, 20, 30, 40, 50, 60]`, what will `nums[1:4]` return?

- A) `[10, 20, 30]`
- B) `[20, 30, 40]`
- C) `[20, 30, 40, 50]`
- D) `[10, 20, 30, 40]`

2. Which of the following expressions correctly reverses a list `lst` using slicing?

- A) `lst[0:-1]`
- B) `lst[-1:0]`
- C) `lst[::-1]`
- D) `lst[:]`

3. What is the output of the following Python code snippet?

```
letters = ['a', 'b', 'c', 'd', 'e']  
print(letters[2:])
```

- A) `['a', 'b']`
- B) `['c', 'd', 'e']`
- C) `['b', 'c', 'd', 'e']`
- D) `['c', 'd']`

4. What will be the output of the following slice operation?

```
data = [1, 2, 3, 4, 5, 6, 7, 8]  
print(data[1:7:2])
```

- A) `[2, 4, 6]`
- B) `[1, 3, 5, 7]`
- C) `[2, 5, 8]`
- D) `[2, 4, 6, 8]`

5. Consider the statement: `mystri = [5, 10, 15, 20]`. What does `mystri[:-1]` yield?

- A) `[20]`
- B) `[5, 10, 15]`
- C) `[10, 15, 20]`
- D) `[20, 15, 10]`

6. If you execute `my_list.remove(x)`, what happens if the value `x` appears multiple times in `my_list`?

- A) All occurrences of `x` are removed.
- B) Only the last occurrence of `x` is removed.
- C) Only the first occurrence of `x` is removed.
- D) A `ValueError` is raised because duplicates are ambiguous.

7. What happens if you call `.remove(100)` on a list that does not contain the element `100`?

- A) The list remains unchanged and no error is raised.
- B) It returns `None` and outputs a warning.
- C) It raises a `ValueError`.
- D) It raises an `IndexError`.

8. Which list method removes and returns the last element of a list by default?

- A) `remove()`
- B) `delete()`
- C) `pop()`
- D) `discard()`

9. What will be the output of the following code snippet?

```
fruits = ['apple', 'banana', 'cherry']
result = fruits.pop(1)
print(fruits, result)
```

- A) `['apple', 'cherry']` `banana`
- B) `['banana', 'cherry']` `apple`
- C) `['apple', 'banana']` `cherry`
- D) `['apple', 'cherry']` `1`

10. How do you completely delete all items from a list named `items`, leaving it completely empty?

- A) `items.remove_all()`
- B) `items.clear()`
- C) `del items`
- D) `items.clean()`

11. What is the main difference between the `del` statement and the `.remove()` method?

- A) `del` deletes by index, whereas `remove()` deletes by value.
- B) `del` deletes by value, whereas `remove()` deletes by index.
- C) `del` can only delete whole lists, while `remove()` deletes individual items.
- D) There is no difference; they are completely interchangeable.

12. What is the output of the following code?

```
arr = [10, 20, 30, 40]
del arr[1:3]
print(arr)
```

- A) `[10, 40]`
- B) `[10, 20, 40]`
- C) `[40]`
- D) `[10, 30, 40]`

13. Which list method adds an item to the end of the list?

- A) `insert()`
- B) `extend()`
- C) `append()`
- D) `push()`

14. What will be the final state of list `x` after running this code?

```
x = [1, 2]
x.extend([3, 4])
print(x)
```

- A) `[1, 2, [3, 4]]`
- B) `[1, 2, 3, 4]`
- C) `[3, 4, 1, 2]`
- D) `[1, 2, 7]`

15. What does `items.insert(0, 'A')` do on a non-empty list `items`?

- A) Replaces the element at index 0 with 'A'.
- B) Adds 'A' at the very beginning of the list, shifting existing items to the right.
- C) Appends 'A' to the end of the list.
- D) Raises an `IndexError`.

16. What will be the output of the following sequence?

```
vowels = ['e', 'a', 'u', 'o', 'i']
vowels.sort()
print(vowels)
```

- A) ['e', 'a', 'u', 'o', 'i']
- B) ['i', 'o', 'u', 'a', 'e']
- C) ['a', 'e', 'i', 'o', 'u']
- D) None

17. How does the list method `.sort()` differ from the built-in function `sorted()`?

- A) `sort()` modifies the original list in-place; `sorted()` returns a new sorted list.
- B) `sorted()` modifies the original list; `sort()` returns a new list.
- C) `sort()` can sort strings, while `sorted()` can only sort numbers.
- D) They operate in exactly the same way and take identical execution times.

18. What is the output of the following code block?

```
numbers = [1, 2, 3, 2, 4, 2]
print(numbers.count(2))
```

- A) 1
- B) 2
- C) 3
- D) 5

19. If `values = [10, 20, 30, 40]`, what is returned by `values.index(30)`?

- A) 2
- B) 3
- C) 1
- D) [10, 20]

20. What is the slice outcome of `my_list[5:2]` if `my_list` contains 10 items?

- A) A list containing elements from index 2 to 5 in reverse.
- B) An empty list `[]`.
- C) An `IndexError` exception.
- D) A list containing the first 3 elements.